

MUKUL KUMAR TYAGI

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SUMMARY

Data scientist working at the intersection of forecasting, financial planning, and workforce analytics. I build the forecasts and reporting that budget, headcount, and sourcing decisions run on — currently on a Stellantis engagement, where I automate financial and resource-planning reporting across SAP, Airflow, and Power BI. Strong in Python and SQL, with recent depth in experimentation and causal inference (see portfolio).

SKILLS

Forecasting & statistics: SARIMA/SARIMAX time series, regression, rolling-origin backtesting, A/B test design, power analysis, causal inference (difference-in-differences, synthetic control), survival analysis

Programming & data: Python (pandas, NumPy, scikit-learn, statsmodels, PySpark), SQL (CTEs, window functions), Excel VBA

Platforms: Snowflake, Databricks, Spark, Hadoop, Airflow, SAP, AWS SageMaker

Reporting: Power BI, Tableau, Streamlit, executive dashboards and written analysis

Domains: Budget and CapEx forecasting, FTE and resource planning, outsourcing and supplier cost analysis, HR/workforce data, demand planning

EXPERIENCE

Data Analyst · Segula Technologies (client: Stellantis) — Auburn Hills, MI Oct 2025 – Present

- Deliver workload and financial reporting analytics that support budget forecasts, resource planning, and outsourcing efficiency across engineering functions.
- Build scenario-based forecasts for workforce and capacity planning, giving Finance and CapEx teams a view of headcount cost under different demand assumptions.
- Automate recurring financial dashboards and reporting pipelines in Python, SQL, Power BI, and Streamlit, replacing manual reporting cycles.
- Validate data across Airflow, Resource Planning (SAP), and finance systems to keep reporting reconciled with CapEx budgets.
- Produce supplier portfolio risk and performance analytics used to identify cost optimisation opportunities in the Outsourcing Efficiency Drive.
- Consolidate reporting for leadership reviews, translating operational detail into the few numbers an investment decision turns on.

Decision Scientist · Alfamodo Lifestyle — Remote Aug 2025 – Feb 2026

- Ran A/B tests and causal analyses to evaluate marketing spend and campaign ROI.
- Built predictive and behavioural models for customer retention and segmentation.
- Worked with engineering to build data pipelines and client performance dashboards in Python and SQL.
- Presented findings and recommendations directly to executive stakeholders.

Business Intelligence Analyst (Capstone) · Raymond Corporation (Toyota Industries) Jan 2025 – Jun 2025

- Built SARIMA/SARIMAX demand forecasts incorporating CPI and interest rates as exogenous drivers, raising forecast accuracy from 82% to 96% on the project dataset (synthetic data built to mirror the client's structure).
- Built SQL and Python pipelines to assemble forecasting inputs so the process could be rerun each planning cycle.
- Audited existing forecasting tools with Sales and Supply Chain teams and closed gaps between legacy reports and current business requirements.
- Translated 30+ regression outputs into Tableau and Power BI dashboards with user guides for pricing, supply chain, and production planning.
- Delivered two weeks ahead of schedule, giving leadership earlier visibility into market trends.

Programmer Analyst · Cognizant Technology Solutions — Remote Mar 2021 – Jun 2024

- Built ETL pipelines in Python, SQL, and Hadoop processing millions of insurance records weekly for actuarial and financial reporting.
- Led requirements gathering, functional specification, and change control for digital transformation work in the insurance domain.
- Designed and optimised data models for customer and claims workflows.

- Automated reconciliation and reporting tasks with Excel VBA and Power BI, cutting manual effort.
- Coordinated UAT and QA for production releases, and supported IT audits covering data governance and control gaps.

SELECTED PROJECTS

Portfolio of seven experimentation and forecasting projects — mukultyagi481.github.io

- Forecasting on real data: backtested SARIMA, ETS, and naive benchmarks over 78 origins on 30 years of published price data; no model beat the random walk at a statistically detectable level — reported as-is, with the planning implications.
- Experimentation methods: showed pre-experiment covariate adjustment (CUPED) cuts required sample size by 26%, and that daily peeking raises the false positive rate from 5% to 22%.
- SQL analytics: cohort retention, market-level ranking, value concentration, and reconciliation checks using CTEs and window functions.

EDUCATION

MS, Business Analytics (STEM) · University at Buffalo, SUNY	Jun 2025
Statistical analytics, predictive analytics, database management, supply chain analytics and modelling.	
Advanced Certificate, Business and Data Analytics · Indian Institute of Management	Jun 2024
B.Tech, Computer Science and Engineering · Lovely Professional University	Jul 2021

CERTIFICATIONS

McKinsey Forward Program · Microsoft Certified: Azure Fundamentals